Unit 9: Power Rule & Basic Rules — Full Solutions

Warm-up

1. $y = x^5$

Bring the exponent 5 down front as a multiplier, then subtract 1 from the exponent:

$$y' = 5x^{5-1} = 5x^4$$

2. $y = x^{10}$

$$y' = 10x^{10-1} = 10x^9$$

3. $y = 7x^3$

The 7 just tags along; apply the power rule to x^3:

$$y' = 7 \cdot 3x^2 = 21x^2$$

4. $y = 4x^2 - 3x + 5$

Differentiate term by term. $4x^2 \to 8x$. $-3x \to -3$ (since $x = x^1$, and $1 \cdot x^0 = 1$). $5 \to 0$ (constant).

$$y' = 8x - 3$$

5. $y = x^3 - 2x^2 + x - 9$

$x^3 \to 3x^2$. $-2x^2 \to -4x$. $x \to 1$. $-9 \to 0$.

$$y' = 3x^2 - 4x + 1$$

6. $y = -3x^4 + 2x^2$

$-3x^4 \to -12x^3$. $2x^2 \to 4x$.

$$y' = -12x^3 + 4x$$

7. $y = 10$

A plain constant, no x attached — its derivative is 0.

$$y' = 0$$

8. $y = 6x$

Think of this as $6x^1$: bring down the 1, then the exponent becomes 0, and $x^0 = 1$.

$$y' = 6 \cdot 1 \cdot x^0 = 6$$

Standard

9. $y = \sqrt{x}$

Rewrite as a power first: $y = x^{1/2}$.

$$y' = \frac{1}{2}x^{\frac{1}{2}-1} = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$$

10. $y = \dfrac{1}{x^2}$

Rewrite: $y = x^{-2}$.

$$y' = -2x^{-2-1} = -2x^{-3} = -\frac{2}{x^3}$$

11. $y = \dfrac{1}{x^3}$

Rewrite: $y = x^{-3}$.

$$y' = -3x^{-3-1} = -3x^{-4} = -\frac{3}{x^4}$$

12. $y = 3\sqrt{x} - \dfrac{2}{x}$

Rewrite: $y = 3x^{1/2} - 2x^{-1}$.

Differentiate each term: $3x^{1/2} \to 3 \cdot \frac{1}{2}x^{-1/2} = \frac{3}{2}x^{-1/2}$. $-2x^{-1} \to -2 \cdot (-1)x^{-2} = 2x^{-2}$.

$$y' = \frac{3}{2}x^{-1/2} + 2x^{-2} = \frac{3}{2\sqrt{x}} + \frac{2}{x^2}$$

13. $y = x^{3/2}$

$$y' = \frac{3}{2}x^{\frac{3}{2}-1} = \frac{3}{2}x^{1/2} = \frac{3}{2}\sqrt{x}$$

14. $y = \dfrac{4}{\sqrt{x}}$

Rewrite: $y = 4x^{-1/2}$.

$$y' = 4 \cdot \left(-\frac{1}{2}\right) x^{-1/2-1} = -2x^{-3/2} = -\frac{2}{x\sqrt{x}}$$

15. $y = 2x^3 - 5\sqrt{x} + \dfrac{1}{x}$

Rewrite: $y = 2x^3 - 5x^{1/2} + x^{-1}$.

Differentiate each term: $2x^3 \to 6x^2$. $-5x^{1/2} \to -5 \cdot \frac{1}{2}x^{-1/2} = -\frac{5}{2}x^{-1/2}$. $x^{-1} \to -1 \cdot x^{-2} = -x^{-2}$.

$$y' = 6x^2 - \frac{5}{2}x^{-1/2} - x^{-2} = 6x^2 - \frac{5}{2\sqrt{x}} - \frac{1}{x^2}$$

Challenge

16. $y = (x^2+1)^2$

Expand first: $(x^2+1)^2 = (x^2+1)(x^2+1) = x^4 + 2x^2 + 1$.

Now differentiate term by term: $x^4 \to 4x^3$. $2x^2 \to 4x$. $1 \to 0$.

$$y' = 4x^3 + 4x$$

17. $y = x(x^2-3)$

Expand first: $x(x^2-3) = x^3 - 3x$.

$$y' = 3x^2 - 3$$

18. $y = \dfrac{x^3 + 2x^2 - 5x}{x}$

Divide every term in the top by x first (valid wherever $x \neq 0$):

$$\frac{x^3}{x} + \frac{2x^2}{x} - \frac{5x}{x} = x^2 + 2x - 5$$

Now differentiate:

$$y' = 2x + 2$$

19. $y = \dfrac{2x^4 - 3x^2 + x}{x^2}$

Divide every term in the top by x^2 first:

$$\frac{2x^4}{x^2} - \frac{3x^2}{x^2} + \frac{x}{x^2} = 2x^2 - 3 + x^{-1}$$

Now differentiate: $2x^2 \to 4x$. $-3 \to 0$. $x^{-1} \to -x^{-2}$.

$$y' = 4x - x^{-2} = 4x - \frac{1}{x^2}$$

20. $y = (2x-1)^2$

Expand first: $(2x-1)^2 = (2x-1)(2x-1) = 4x^2 - 4x + 1$.

$$y' = 8x - 4$$

21. Find all x where the tangent to $y = x^3 - 3x$ is horizontal.

First find the derivative:

$$y' = 3x^2 - 3$$

A horizontal tangent happens where the slope is 0:

$$3x^2 - 3 = 0$$

$$3x^2 = 3$$

$$x^2 = 1$$

$$x = \pm 1$$

Answer: the tangent line is horizontal at $x = 1$ and at $x = -1$.

Applied

22. $s(t) = t^3 - 6t^2 + 9t$

Differentiate to get the velocity function:

$$v(t) = s'(t) = 3t^2 - 12t + 9$$

Evaluate at $t = 1$:

$$v(1) = 3(1) - 12(1) + 9 = 3 - 12 + 9 = 0$$

Answer: $v(t) = 3t^2 - 12t + 9$, and $v(1) = 0$. At exactly $t = 1$ second, the object has momentarily stopped moving — likely turning around at that instant.

23. $C(x) = 0.01x^3 - 0.6x^2 + 13x + 100$

Differentiate to get the marginal cost function:

$$C'(x) = 0.03x^2 - 1.2x + 13$$

Evaluate at $x = 10$:

$$C'(10) = 0.03(100) - 1.2(10) + 13 = 3 - 12 + 13 = 4$$

Answer: $C'(x) = 0.03x^2 - 1.2x + 13$, and $C'(10) = 4$. At a production level of 10 units, each additional unit costs about $4 more to make.

24. $A(x) = x(20 - x)$

First expand: $A(x) = 20x - x^2$.

Differentiate:

$$\frac{dA}{dx} = 20 - 2x$$

Set it equal to 0:

$$20 - 2x = 0$$

$$x = 10$$

Answer: $\frac{dA}{dx} = 20 - 2x$, and the area's rate of change is 0 at $x = 10$. This is the value of x that makes the flat tangent line occur on the area graph — meaning it's a candidate for the largest possible garden area (we'll build the full tools to confirm this is a true maximum in a later unit).

25. $R(x) = 50x - 0.02x^2$

Differentiate:

$$R'(x) = 50 - 0.04x$$

Evaluate at $x = 100$:

$$R'(100) = 50 - 0.04(100) = 50 - 4 = 46$$

Answer: $R'(x) = 50 - 0.04x$, and $R'(100) = 46$. At a production level of 100 units, selling one more unit adds about $46 to total revenue.